

Boulhit Walid

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EDUCATION

ENSA Berrechid, Hassan 1st University <i>Engineering Degree in Aeronautical Engineering (Embedded Systems)</i>	Berrechid, Morocco 2025 – 2028
– Relevant Coursework: CAD (Catia V5), Signal Processing, Thermodynamics, Heat Transfer, C++, Python. – Member of Aero Club ENSAB: Active participation in technical workshops and drone design competitions.	
ENSA Berrechid, Hassan 1st University <i>Preparatory Cycle (Maths, Physics, Engineering Science)</i>	Berrechid, Morocco 2023 – 2025

PROJECTS

Boeing 737-9 MAX Surface Modeling <i>Catia V5 (GSD/Assembly), Aerospace Design</i>	Mars 2026
– Advanced Surfacing: Full 3D modeling of the fuselage and wings using Generative Shape Design (GSD) to ensure aerodynamic continuity (Class-A surfaces). – Assembly Management: Managed a large-scale assembly hierarchy, integrating landing gear and empennage structures with precise constraints. – Reverse-engineered structural components based on technical blueprints to validate geometric accuracy.	
Radial Engine & V8 Mechanism Design <i>Catia V5 (Part/Kinematics), Mechanics</i>	Jan 2026
– Mechanical Design: Detailed parametric modeling of engine components (pistons, connecting rods, crankshaft) using Part Design. – Kinematic Simulation: Simulated the combustion cycle movement using DMU Kinematics to detect collisions and validate clearance. – Generated 2D technical drawings (Drafting) with GD&T standards for manufacturing specifications.	
Thermal Dissipation Analysis <i>Python, Thermodynamics, Numerical Methods</i>	Nov 2025
– Developed a numerical solver in Python to model heat transfer (Conduction/Convection) in engine cooling fins. – Simulated steady-state temperature distribution across cylindrical geometries using Fourier's Law. – Visualized thermal gradients to optimize material selection for high-temperature aeronautical applications.	

TECHNICAL SKILLS

CAD & Simulation: Catia V5 (Expertise in Part Design, GSD, Assembly, Drafting), DMU Kinematics, Proteus.
Engineering Science: Thermodynamics, Heat Transfer, Signal Processing (FFT, Filtering), Mechanics of Flight.
Programming: C/C++ (Embedded), Python (Scientific Computing: NumPy, Pandas), Arduino, MATLAB.
Development Tools: Git/GitHub, LaTeX, React.js (used for telemetry visualization).

EXPERIENCE

ALX Web Development Program <i>ALX, African Leadership Group</i>	2023 – 2024 Remote
– Collaborated in agile teams to deliver complex technical projects under strict deadlines. – Developed rigorous documentation and version control habits (Git), transferable to industrial quality standards. – Gained proficiency in data structuring and UI logic, enhancing ability to design intuitive control interfaces.	

CERTIFICATIONS

Python for Everybody Specialization (Univ. of Michigan) – *Data Structures & Algorithms*
Meta Front-End Developer Certificate – *Project Management & UI Design*